



PREMIUM STUDY GUIDE

MASTER EXCEL IN 30 DAYS

A Complete Day-by-Day Course
for Learners & Teachers

Beginner to Advanced | Formulas · Functions · Charts · PivotTables ·
Automation

Includes worked examples, practice exercises, and a capstone project

30

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"Explaining a concept is the fastest way to master it."

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WEEK 1

Excel Foundations

Days 1-7: Learn the Excel interface, enter and format data, and master your first formulas and functions.

Why This Week Matters

Every advanced Excel skill — PivotTables, macros, dashboards — is just this week's ideas combined in bigger ways. A shaky grasp of cell references or the equals sign doesn't disappear as the course progresses; it resurfaces as confusing errors weeks later. Students (and teachers) who feel tempted to rush through Week 1 because it "seems easy" are exactly the ones who benefit most from slowing down here.

This Week's Lessons

DAY 1	Getting to Know Excel	Interface, cells, workbook vs worksheet
DAY 2	Entering & Editing Data	Data types, AutoFill, Flash Fill
DAY 3	Basic Formulas & Cell References	Operators, relative vs absolute (\$)
DAY 4	Formatting Cells	Number formats, conditional formatting
DAY 5	Rows, Columns & Sheets	Freeze panes, multiple sheets
DAY 6	Basic Functions	SUM, AVERAGE, MIN, MAX, COUNT
DAY 7	Week 1 Review	Mini project: personal budget

By the End of This Week, You'll Be Able To

- Navigate a workbook confidently using the Name Box, keyboard shortcuts, and multiple sheets
- Write a formula from scratch and correctly choose between relative and absolute (\$) references
- Format a table so that both numbers and highlights communicate meaning at a glance
- Use SUM, AVERAGE, MIN, MAX, and COUNT to summarize a column of numbers
- Explain the difference between a workbook, a worksheet, a cell, and a range to someone new to Excel

📌 Teacher's Note

If you're teaching this week to a group, resist the urge to move fast just because the material looks simple on the surface. The habits formed in these first seven days — checking cell references before copying a formula, glancing at data alignment to catch entry mistakes — are the same habits that prevent hours of debugging in Week 3 and Week 4. Slow is smooth, and smooth is fast.

Getting to Know Excel

📌 Teacher's Note

Students almost always want to jump straight to formulas, but skipping the mental model of rows, columns, and cell addresses causes confusion for weeks afterward. I spend real time here on purpose: once someone truly understands that a cell is simply an intersection of a row and a column — and that every single feature in Excel is built on top of that one idea — everything else in this course becomes dramatically easier to teach and to learn. Don't rush Day 1, even if it feels basic.

What Is Excel?

Microsoft Excel is a **spreadsheet program** used to organize, calculate, and analyze data using rows and columns. A file is called a **workbook**, and each page inside it is called a **worksheet** (or "sheet"). Every worksheet is made of a grid of **cells**, and every cell has an address made from its **column letter** and **row number** — for example, cell **B3** is in column B, row 3.

The Excel Interface

- **Ribbon** — the toolbar at the top (Home, Insert, Formulas, Data, View tabs)
- **Formula Bar** — shows the content/formula of the selected cell
- **Name Box** — shows the address of the selected cell, and lets you jump to any cell
- **Sheet Tabs** — at the bottom, let you switch between worksheets
- **Cell Reference** — the unique address of a cell, e.g. A1, C5, Z100

Workbook vs Worksheet vs Cell

Term	What it means	Example
Workbook	The entire Excel file	Sales_2026.xlsx
Worksheet	One page/tab inside the workbook	Sheet1, "Jan Sales"
Cell	One box in the grid	A1, D12
Range	A group of cells	A1:D10

💡 **TIP:** Press **Ctrl + N** to open a new workbook and **Ctrl + S** to save it at any time.

How to Navigate

Action	How to do it
Move one cell	Arrow keys
Jump to last used cell	Ctrl + End
Jump to A1	Ctrl + Home
Select a range	Click and drag, or Shift + Arrow keys
Go to a specific cell	Type address into the Name Box and press Enter

👉 **PRACTICE:** Open a new workbook. Rename "Sheet1" to "MyFirstSheet". Click on cell D4 using the Name Box, then type your name into it.

Quick Check

What is the address of the cell in column C, row 7?
Answer: C7

Entering & Editing Data

📌 Teacher's Note

The distinction between how data is *entered* and how it's *stored* trips up beginners constantly — for example, typing "01/05" might become a date you didn't intend. Teach students to glance at the alignment of what they just typed (numbers snap right, text snaps left) as an instant, built-in check that Excel understood their entry correctly. This one habit prevents a huge share of downstream formula errors.

Types of Data in Excel

Excel automatically detects what kind of data you type. Understanding this helps you avoid formatting mistakes.

Data Type	Example	Default Alignment
Text (label)	Product Name	Left
Number	1500	Right
Date	23-08-2026	Right
Currency	\$1,500.00	Right
Formula	=A1+A2	Right (shows result)

Entering & Editing Data

- Click a cell and start typing, then press **Enter** (moves down) or **Tab** (moves right)
- Double-click a cell (or press **F2**) to edit its existing content
- Press **Delete** to clear a cell's content
- **Esc** cancels an entry you're in the middle of typing

AutoFill - Excel's Time Saver

Drag the small square at the bottom-right of a cell (the **fill handle**) to copy a value or continue a pattern automatically.

You type in A1:A2	Drag down fills with
1, 2	3, 4, 5, 6 ...
Monday	Tuesday, Wednesday, Thursday ...
Jan	Feb, Mar, Apr ...
Q1	Q2, Q3, Q4 ...

Flash Fill (Ctrl + E)

Flash Fill detects a pattern from what you type and fills the rest of the column instantly.

Column A (Full Name)	Column B (you type)	Flash Fill result
Amit Sharma	Amit	(press Ctrl+E → fills First Names down)
Priya Verma	Priya	Priya
Rohan Gupta		Rohan

💡 **TIP:** Type the first 1-2 examples, select the column, then press **Ctrl + E** — Excel completes the rest based on your pattern.

🔗 **PRACTICE:** In a new sheet, type the months Jan-Dec using AutoFill in column A. In column B, type 3 full names and use Flash Fill to extract just the first names in column C.

WEEK 1 — Foundations • DAY 3 OF 30

Basic Formulas & Cell References

📌 Teacher's Note

The equals sign is the single most important keystroke in this entire course — it's the difference between Excel treating your entry as inert text and treating it as a live instruction to calculate. Absolute references (\$) are conceptually simple but procedurally awkward for new learners, because the mistake often doesn't show up until several rows later. I recommend having students deliberately copy a relative-reference formula down first, watch it break, and only then introduce the \$ as the fix — the error makes the lesson stick far better than explaining the rule up front.

Why Formulas Matter

Every formula in Excel starts with an **equals sign (=)**. Formulas let Excel calculate automatically instead of you doing math by hand.

Basic Arithmetic Operators

Operator	Meaning	Example	Result
+	Addition	=5+3	8
-	Subtraction	=10-4	6
*	Multiplication	=6*7	42
/	Division	=20/4	5
^	Exponent (power)	=2^3	8

Using Cell References Instead of Numbers

Instead of typing raw numbers, refer to the cells that hold them — this way, if the data changes, the formula's result updates automatically.

Cell	Content
A1	200
A2	50
A3	=A1+A2 → 250

```
=A1+A2 // adds the values in A1 and A2
```

Relative vs Absolute References — The #1 Beginner Mistake

When you copy a formula to other cells, Excel shifts **relative** references automatically, but keeps **absolute** references fixed using the **\$** sign.

Reference type	Looks like	What happens when copied down
Relative	A1	Changes to A2, A3, A4 ...
Absolute (locked)	\$A\$1	Stays \$A\$1 always
Mixed (column locked)	\$A1	Column fixed, row changes
Mixed (row locked)	A\$1	Row fixed, column changes

⚠ **WATCH OUT:** Forgetting the \$ sign when a formula needs to reference one fixed cell (like a tax rate) is the most common formula error beginners make.

Worked Example: Applying a Fixed Tax Rate

	A: Item	B: Price	C: Tax (uses \$E\$1)
1	Pen	10	=B1*\$E\$1
2	Book	200	=B2*\$E\$1
3	Bag	500	=B3*\$E\$1
		E1 = Tax Rate →	0.05 (5%)

Copying **C1** down to C2 and C3 keeps referring to **\$E\$1** (the tax rate) while B1 automatically becomes B2, then B3.

🔗 **PRACTICE:** Create a small price list with 5 items. Add a fixed "Discount %" cell, and write one formula using a mixed reference (\$) that you can copy down for all 5 items.

📌 Teacher's Note

Formatting is often treated as cosmetic, but it's really a communication skill: the format of a number tells the reader what kind of thing it is before they've even processed the value itself. A well-placed color scale can convey a trend that would otherwise take a full paragraph to explain. When teaching, frame formatting decisions as "what does my reader need to notice first?" rather than "what looks nice?" — it produces far more disciplined spreadsheets.

Why Formatting Matters

Formatting changes how data *looks* without changing the actual value stored in the cell. Good formatting makes spreadsheets easier to read and teach from.

Number Formats

Raw Value	Format Applied	Displayed As
1500	Number (2 decimals)	1500.00
1500	Currency	\$1,500.00
0.25	Percentage	25%
45123	Date	08/23/2026
1500000	Number with commas	1,500,000

Apply these from **Home tab → Number group**, or press **Ctrl + 1** to open Format Cells for full control.

Font, Color & Borders

- **Bold / Italic / Underline:** Ctrl+B, Ctrl+I, Ctrl+U
- **Fill color:** Home → Fill Color bucket icon — great for highlighting headers
- **Borders:** Home → Borders dropdown, or Ctrl+1 → Border tab for custom styles
- **Merge & Center:** useful for titles spanning multiple columns

Conditional Formatting — Make Data Highlight Itself

Conditional formatting automatically changes a cell's appearance based on its value.

Student	Score	Rule Applied
Aisha	92	Green if ≥ 80
Ravi	65	Orange if 50-79
Meera	38	Red if < 50

Path: **Home → Conditional Formatting → Highlight Cell Rules / Color Scales / Data Bars**.

🔗 **TIP:** Use **Data Bars** for a quick visual comparison between numbers in a column — no chart needed.

🔗 **PRACTICE:** Format a small list of 5 test scores: apply Currency format is not relevant here, so instead apply Conditional Formatting so scores below 50 turn red and scores 80+ turn green.

WEEK 1 — Foundations • DAY 5 OF 30

Rows, Columns, Sheets & Freeze Panes

📌 Teacher's Note

Multi-sheet workbooks are where students first start to think about data architecture rather than just data entry — deciding what belongs on its own sheet versus what belongs alongside other information is a genuinely transferable skill, useful well beyond Excel. Freeze Panes seems like a minor convenience feature, but for anyone reviewing a 500-row sheet, losing the header row is a real source of costly mistakes, so I treat it as a habit to build early rather than an optional trick.

Rows & Columns: Insert, Delete, Resize

Task	How
Insert a row	Right-click row number → Insert
Insert a column	Right-click column letter → Insert
Delete a row/column	Right-click → Delete
Resize column width	Drag the border between column letters, or double-click to auto-fit
Hide a row/column	Right-click → Hide (Unhide to bring back)

Freeze Panes — Keep Headers Visible

When scrolling through long data, freezing panes keeps your header row (or first column) always visible.

Path: **View tab → Freeze Panes → Freeze Top Row** (or Freeze First Column).

 **TIP:** For a header + ID column both frozen at once, click the cell just below and right of what you want frozen (e.g. B2), then choose Freeze Panes.

Working with Multiple Sheets

- Add a sheet: click the + next to the sheet tabs
- Rename: double-click the sheet tab name
- Move/copy: right-click tab → Move or Copy
- Reference another sheet in a formula: **=Sheet2!A1**

```
=SUM(Sheet2!A1:A10) // adds values from a range on a different sheet
```

Grouping & Outline

Select multiple rows or columns → **Data tab → Group** to create collapsible sections — useful for hiding details in a report while teaching a summary view first.

 **PRACTICE:** Create 2 sheets: "Data" and "Summary". In Data, enter 10 rows of sample sales figures. In Summary, write a formula that pulls a total from the Data sheet using SUM(Data!...).

WEEK 1 — Foundations • DAY 6 OF 30

Basic Functions: SUM, AVERAGE, MIN, MAX, COUNT

 **Teacher's Note**

SUM and AVERAGE look almost too simple to dwell on, but this is exactly where I introduce the idea of a function's *argument* — the input it needs to do its job — because every advanced function later in this course (VLOOKUP, IF, INDEX/MATCH) is built from that same basic grammar: FUNCTION(argument, argument, ...). A student who deeply understands SUM(range) already understands the syntax shape of every function they'll meet for the rest of the course.

The 5 Functions Every Beginner Must Know

A **function** is a ready-made formula. All functions follow this pattern: **=FUNCTION(argument1, argument2, ...)**

Function	Purpose	Example	Result
SUM	Adds numbers	=SUM(A1:A5)	Total of A1 to A5
AVERAGE	Finds the mean	=AVERAGE(A1:A5)	Average of A1 to A5
MIN	Smallest value	=MIN(A1:A5)	Lowest number
MAX	Largest value	=MAX(A1:A5)	Highest number
COUNT	Counts numeric cells	=COUNT(A1:A5)	How many contain numbers

Worked Example: Class Test Scores

Student	Score
Aisha	88
Ravi	72
Meera	95
Karan	60
Total →	=SUM(B2:B5) = 315
Average →	=AVERAGE(B2:B5) = 78.75
Highest →	=MAX(B2:B5) = 95
Lowest →	=MIN(B2:B5) = 60

AutoSum Shortcut

Click an empty cell below/right of a number range, then press **Alt + =** — Excel guesses the range and inserts SUM automatically.

COUNTA and COUNTBLANK — Bonus Functions

Function	Counts
COUNTA	Non-empty cells (text or numbers)
COUNTBLANK	Empty cells in a range

⚠ WATCH OUT: SUM and AVERAGE ignore text and blank cells automatically — they only calculate numeric values.

➡ **PRACTICE:** Build a table of 6 students' marks in 3 subjects. Use SUM to get each student's total, AVERAGE for their percentage, and MAX/MIN to find the topper and lowest scorer overall.

WEEK 1 — Foundations • DAY 7 OF 30

Week 1 Review + Mini Project

📌 Teacher's Note

Review days matter more than they get credit for. In my classroom, the mini-project at the end of each week is graded more heavily than any single day's practice exercise, because it's the first place students have to decide, unprompted, which tool solves which problem — that decision-making is the actual skill we're building, not memorization of individual functions.

Week 1 Recap

Day	Skill
1	Excel interface: ribbon, cells, workbook vs worksheet
2	Entering data, AutoFill, Flash Fill
3	Basic formulas, relative vs absolute references (\$)
4	Number formatting & conditional formatting
5	Rows/columns, freeze panes, multiple sheets
6	SUM, AVERAGE, MIN, MAX, COUNT

Mini Project: Personal Monthly Budget

Build a one-sheet budget tracker using everything learned this week.

Category	Budgeted	Actual	Difference
Rent	15000	15000	=B2-C2
Groceries	6000	6800	=B3-C3
Transport	2000	1750	=B4-C4
Entertainment	1500	2100	=B5-C5
Total	=SUM(B2:B5)	=SUM(C2:C5)	=SUM(D2:D5)

Project Checklist

- ☐ Apply Currency format to money columns
- ☐ Bold and color-fill the header row
- ☐ Add conditional formatting: red if Actual > Budgeted
- ☐ Freeze the header row
- ☐ Use SUM for the Total row

🔗 **TIP:** For teaching: have students swap budgets with a partner and find one formatting or formula mistake to fix — this reinforces Week 1 skills through peer review.

WEEK 2

Data Management & Logic

Days 8–14: Sort and filter real datasets, make decisions with IF, and summarize data conditionally.

Why This Week Matters

This is the week Excel stops being a calculator and starts being a decision engine. IF teaches the software to make a choice; COUNTIF and SUMIF teach it to summarize a condition instead of a whole dataset; VLOOKUP teaches it to find one specific answer inside a sea of rows. Individually these feel like small tricks, but together they're the core logic behind almost every real spreadsheet used in a workplace.

This Week's Lessons

DAY 8	Sorting & Filtering Data	Reorder and narrow down data
DAY 9	The IF Function	Logical tests and decisions
DAY 10	COUNTIF, SUMIF & friends	Conditional counting and totals
DAY 11	Advanced Conditional Formatting	Formula-based highlighting
DAY 12	Data Validation	Dropdowns and input rules
DAY 13	The VLOOKUP Function	Look up data from a table
DAY 14	Week 2 Review	Mini project: grade tracker

By the End of This Week, You'll Be Able To

- Sort and filter a dataset without altering the underlying data
- Write single and nested IF statements to automate a decision
- Use COUNTIF/COUNTIFS and SUMIF/SUMIFS to summarize data that meets one or more conditions
- Build a dropdown list with Data Validation to prevent bad data entry
- Use VLOOKUP with FALSE and IFERROR to pull data from another table safely

📌 Teacher's Note

Watch for the moment a student starts combining two of this week's tools without being told to — for example, wrapping a COUNTIF result inside an IF statement. That's the real signal that the underlying logic has clicked, not just the individual function names. I treat that moment as more meaningful than any quiz score.

Sorting & Filtering Data

📌 Teacher's Note

Sorting and filtering feel almost too simple to be worth a full lesson, but they quietly teach a mindset shift that pays off for the rest of the course: that raw data and the *view* of that data are two separate things. Filtering never deletes anything — it only hides rows temporarily — and reinforcing that distinction early prevents a lot of anxious "where did my data go?" moments later on.

Sorting Data

Sorting reorders rows based on the values in one or more columns.

Task	How
Sort A-Z or Z-A	Select data → Home → Sort & Filter, or Data tab → Sort A-Z/Z-A
Sort by multiple columns	Data → Sort → Add Level (e.g. sort by Department, then Salary)
Sort by color/icon	Data → Sort → choose "Cell Color" from Sort On dropdown

Worked Example — Before Sorting

Name	Department	Salary
Ravi	Sales	42000
Aisha	IT	55000
Meera	Sales	48000
Karan	IT	51000

After sorting by **Department (A-Z)** then **Salary (High to Low)**:

Name	Department	Salary
Aisha	IT	55000
Karan	IT	51000
Meera	Sales	48000
Ravi	Sales	42000

Filtering Data

Filtering hides rows that don't match a condition, without deleting any data.

Path: **Data tab → Filter** (or Ctrl+Shift+L) — adds dropdown arrows to your header row.

Filter Type	Example
Text filter	Show only Department = "IT"
Number filter	Show only Salary > 50000
Date filter	Show only rows from "Last Month"

⚠️ **WATCH OUT:** Always click one cell inside your data table before applying Filter, or Excel may only filter part of your range.

💡 **PRACTICE:** Enter a table of 10 employees with Name, Department, and Salary. Sort it by Salary (highest first). Then apply a filter to show only employees earning above 45,000.

The IF Function & Logical Tests

📌 Teacher's Note

IF is the first place students write something that resembles real logic, and it's worth pausing to name that explicitly — a logical test is really just a yes/no question. I've found that students grasp nested IF far faster when they first sketch it out in plain English ("if score is 90 or above, A; otherwise if 75 or above, B; otherwise...") before typing a single character of the formula. Translating English logic into formula syntax is a separate skill from the logic itself, and separating the two steps reduces frustration enormously.

The IF Function — Excel Makes a Decision

IF checks a condition and returns one value if TRUE, another if FALSE.

```
=IF(logical_test, value_if_true, value_if_false)
```

Worked Example: Pass or Fail

Student	Score	Result
Aisha	82	=IF(B2>=40,"Pass","Fail") → Pass
Ravi	35	=IF(B3>=40,"Pass","Fail") → Fail
Meera	40	=IF(B4>=40,"Pass","Fail") → Pass

Comparison Operators (used inside logical tests)

Operator	Meaning
=	Equal to
>	Greater than
<	Less than
>=	Greater than or equal to
<=	Less than or equal to
<>	Not equal to

Nested IF — More Than 2 Outcomes

You can place one IF inside another to test multiple conditions (grades A/B/C/F, for example).

```
=IF(B2>=90,"A",IF(B2>=75,"B",IF(B2>=50,"C","F")))
```

Score	Grade	Formula Result
95	A	
80	B	
60	C	
30	F	

AND / OR with IF

```
=IF(AND(B2>=40,C2>=40),"Pass","Fail") // both subjects must pass
```

```
=IF(OR(B2="VIP",C2>10000),"Priority","Standard")
```

💡 **TIP:** Excel 365 also has a cleaner alternative called **IFS**, covered later this week, for many conditions without deep nesting.

🔗 **PRACTICE:** Build a table of 6 students with Math and Science scores. Use a nested IF (or AND) to mark a student "Pass" only if BOTH scores are 40 or above.

📌 Teacher's Note

COUNTIF and SUMIF are usually the moment a class has its first "aha" about Excel's real power — instead of manually filtering and adding, one formula does both at once. It's worth explicitly connecting these back to the IF function from the previous day: SUMIF is essentially "IF plus SUM" bundled together. Naming that relationship helps students build a mental map of how functions relate to each other rather than seeing each as an isolated fact to memorize.

COUNTIF — Count Cells That Match a Condition

```
=COUNTIF(range, criteria)
```

Region	Sales
North	4500
South	3200
North	5100
East	2900
North	3900

```
=COUNTIF(A2:A6,"North") // counts 3
```

COUNTIFS — Multiple Conditions

```
=COUNTIFS(A2:A6,"North",B2:B6,">4000") // counts North rows with Sales > 4000 → 2
```

SUMIF — Add Numbers That Match a Condition

```
=SUMIF(A2:A6,"North",B2:B6) // adds Sales only where Region = North → 13500
```

SUMIFS — Add With Multiple Conditions

```
=SUMIFS(B2:B6,A2:A6,"North",B2:B6,">4000") // sums North sales above 4000 → 9600
```

Side-by-Side Summary

Function	Question it answers
COUNTIF	How many rows match ONE condition?
COUNTIFS	How many rows match SEVERAL conditions?
SUMIF	What's the total for rows matching ONE condition?
SUMIFS	What's the total for rows matching SEVERAL conditions?
AVERAGEIF / AVERAGEIFS	Same idea, but calculates an average instead of a total

⚠ **WATCH OUT:** Always put text criteria in quotes: "North", ">4000" — numbers used as a condition also need quotes when combined with an operator.

➡ **PRACTICE:** Using a sales table with Region, Product, and Amount columns (12 rows), write: (1) a COUNTIF for a single region, (2) a SUMIFS for one region AND one product.

WEEK 2 — Data & Logic • DAY 11 OF 30

Advanced Conditional Formatting

📌 Teacher's Note

Formula-based conditional formatting is genuinely more advanced than it looks, because it requires understanding how a rule 'reads' every cell in the selected range relative to one reference point — the mixed-reference thinking from Day 3 resurfaces here in a new context. If a student has forgotten the \$ lock rules, this is the lesson where that gap becomes visible again, so I use it as a diagnostic checkpoint before moving into Week 3.

Conditional Formatting — Beyond the Basics

Day 4 covered simple highlight rules. Today: rules based on formulas and visual data tools.

Data Bars, Color Scales & Icon Sets

Tool	What it shows
Data Bars	In-cell bar proportional to the value — instant visual comparison
Color Scales	Gradient color (e.g. red→yellow→green) based on value size
Icon Sets	Arrows/traffic lights/stars based on value thresholds

Path: **Home** → **Conditional Formatting** → **Data Bars / Color Scales / Icon Sets**

Formula-Based Conditional Formatting

For custom logic, choose **New Rule** → **Use a formula to determine which cells to format**.

```
=C2="Overdue" // highlights the whole row if column C says Overdue
```

Invoice	Due Date	Status
INV001	10-Aug-26	Overdue
INV002	20-Aug-26	Pending
INV003	01-Aug-26	Overdue

Managing Rules

Home → Conditional Formatting → **Manage Rules** lets you edit, reorder, or delete existing rules, and control which rule "wins" when several could apply to the same cell.

 **TIP:** Highlighting a whole row (not just one cell) based on a condition is a favorite classroom demo — it makes patterns in data jump out visually.

 **PRACTICE:** On an invoice list (Invoice No., Due Date, Status), use a formula-based rule to shade the entire row light red wherever Status = "Overdue".

WEEK 2 — Data & Logic • DAY 12 OF 30

Data Validation & Dropdown Lists

 **Teacher's Note**

Data Validation is where I explicitly talk about designing spreadsheets *for other people*, not just for yourself — a shift in mindset that matters enormously once students start building tools that colleagues or classmates will actually use. A dropdown list isn't just a convenience; it's a quiet way of preventing an entire category of future errors before they can happen, and that preventive mindset is worth naming out loud.

Data Validation — Prevent Bad Data Entry

Data Validation restricts what can be typed into a cell — essential for forms, teaching templates, and shared workbooks.

Path: **Data tab** → **Data Validation**

Common Validation Types

Type	Use case	Example setting
Whole number	Age, quantity	between 1 and 100
Decimal	Price, percentage	between 0 and 1
List (dropdown)	Category, Status	North, South, East, West
Date	Deadlines, DOB	between 01/01/2026 and 12/31/2026
Text length	Codes, IDs	exactly 6 characters

Creating a Dropdown List

1. Select the cell(s) you want the dropdown in
2. Data → Data Validation → Allow: **List**
3. Source: type **Yes,No,Maybe** (comma-separated) or select a range like **\$F\$1:\$F\$3**

Task	Status (dropdown)
Prepare slides	▼ Not Started
Grade homework	▼ In Progress
Submit report	▼ Done

Custom Error Alerts

Under the **Error Alert** tab, you can set a custom message (e.g. "Enter a number between 1 and 100") shown when someone breaks the rule.

⚠ **WATCH OUT:** Data Validation only affects new typed entries — it doesn't clean up data that was already in the cells before the rule was applied.

⇒ **PRACTICE:** Create a 5-row task tracker with a Status column restricted to a dropdown list: Not Started, In Progress, Done. Add a custom error message.

WEEK 2 — Data & Logic • DAY 13 OF 30

The VLOOKUP Function

👨 Teacher's Note

VLOOKUP is the function most students have heard of before walking into class, which means it also arrives with the most bad habits already attached — usually forgetting the FALSE argument. I make a point of demonstrating what an approximate match silently returns (a plausible-looking but wrong value) rather than an obvious error, because a formula that fails loudly is far less dangerous than one that fails quietly. That distinction — loud failure versus silent failure — is one of the most important ideas in this entire course.

VLOOKUP — Look Up Data From a Table

VLOOKUP searches for a value in the **first column** of a range, then returns a value from another column in that same row.

```
=VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup])
```

Argument Breakdown

Argument	Meaning
lookup_value	What you're searching for (e.g. a Product ID)
table_array	The full range to search in — lookup column must be first
col_index_num	Which column (counting from 1) to return the answer from
range_lookup	FALSE = exact match (almost always what you want); TRUE = approximate

Worked Example: Product Price Lookup

Product ID	Product Name	Price
P001	Notebook	60
P002	Pen	10
P003	Eraser	5
P004	Sharpener	8

To find the price of P003 from another cell:

```
=VLOOKUP("P003", A2:C5, 3, FALSE) // searches column A, returns column 3 (Price) → 5
```

Common VLOOKUP Errors

Error	Cause	Fix
#N/A	Lookup value not found / typo / mismatched data type	Check spelling, use IFERROR to catch it
#REF!	col_index_num is larger than the table's column count	Recount columns in table_array
Wrong result	Forgot FALSE (defaults to approximate match)	Always add FALSE unless you truly need an approximate match

`=IFERROR(VLOOKUP(A2,PriceList,3,FALSE),"Not Found")` // shows a friendly message instead of #N/A

⚠ WATCH OUT: VLOOKUP can only look to the RIGHT of its search column. If you need to look left, use INDEX & MATCH — covered on Day 19.

👉 **PRACTICE:** Build a 6-item product table (ID, Name, Price). In a separate cell, type a Product ID and use VLOOKUP to automatically display its Name and Price. Wrap it in IFERROR to handle a missing ID.

WEEK 2 — Data & Logic • DAY 14 OF 30

Week 2 Review + Mini Project

📌 Teacher's Note

By the end of Week 2, students have every tool needed to build a genuinely useful piece of software, even if they wouldn't describe it that way. I like to say this explicitly in class: a spreadsheet with logic, lookups, and validation is a small application. Naming it that way tends to noticeably boost confidence and engagement heading into the more visual, PivotTable-heavy content of Week 3.

Week 2 Recap

Day	Skill
8	Sorting & Filtering
9	IF, nested IF, AND/OR
10	COUNTIF(S), SUMIF(S)
11	Advanced conditional formatting
12	Data Validation & dropdowns
13	VLOOKUP + IFERROR

Mini Project: Student Grade Tracker

Name	Math	Science	Average	Grade	Status
Aisha	88	92	<code>=AVERAGE(B2:C2)</code>	Nested IF	<code>=IF(D2>=40,"Pass","Fail")</code>
Ravi	35	45	<code>=AVERAGE(B3:C3)</code>	Nested IF	<code>=IF(D3>=40,"Pass","Fail")</code>

Requirements Checklist

- ☐ Dropdown list for a "Class Section" column (A, B, C) using Data Validation
- ☐ Nested IF to calculate Grade (A/B/C/F) from Average
- ☐ COUNTIF to count how many students passed
- ☐ Conditional formatting: red row for any "Fail" status
- ☐ VLOOKUP cell where a teacher can type a student's name and see their Average instantly

💡 **TIP:** This project combines every Week 2 skill — it works well as a graded classroom assignment since each requirement maps to one function.

WEEK 3

Charts, PivotTables & Lookups

Days 15–21: Turn raw numbers into visual stories with charts and PivotTables, and master INDEX & MATCH plus text functions.

Why This Week Matters

This week marks a shift from calculating answers to communicating them. Week 2 was about getting the right number out of a formula; Week 3 is about making sure another human being — a manager, a classmate, a client — can understand that number in seconds. Charts and PivotTables are where Excel stops being a private calculator and starts being a tool for telling other people something true about the data.

This Week's Lessons

DAY 15	Introduction to Charts	Column, line, and pie charts
DAY 16	Advanced Charts	Combo charts, formatting
DAY 17	PivotTables Basics	Summarize data with drag and drop
DAY 18	PivotTables Advanced	Slicers, grouping, calculated fields
DAY 19	INDEX & MATCH	A more flexible lookup
DAY 20	Text Functions	Cleaning and combining text
DAY 21	Week 3 Review	Mini project: sales dashboard

By the End of This Week, You'll Be Able To

- Choose the right chart type for a given dataset and explain why it's the right choice
- Build a PivotTable to summarize thousands of rows into a clear, small table
- Add slicers and grouped fields to make a PivotTable interactive
- Write an INDEX & MATCH formula that looks up a value in either direction
- Clean messy text data using TRIM, PROPER, and TEXTJOIN

📌 Teacher's Note

This week tends to be the most fun to teach because the payoff is immediate and visual — a messy table becomes a clean PivotTable in seconds, right in front of the class. Use that energy. I deliberately schedule the PivotTable lessons right after a tedious formula-heavy week, because the contrast makes the "wow" moment land even harder.

Introduction to Charts

📌 Teacher's Note

Chart selection is a decision students often make on autopilot — whatever chart type Excel suggests first — so I dedicate real time to the reasoning behind each choice rather than just the mechanics of inserting one. A chart is an argument, not a decoration: it's making a claim about the data, and the wrong chart type can accidentally make a false or misleading claim even when every number in it is correct.

Why Use Charts?

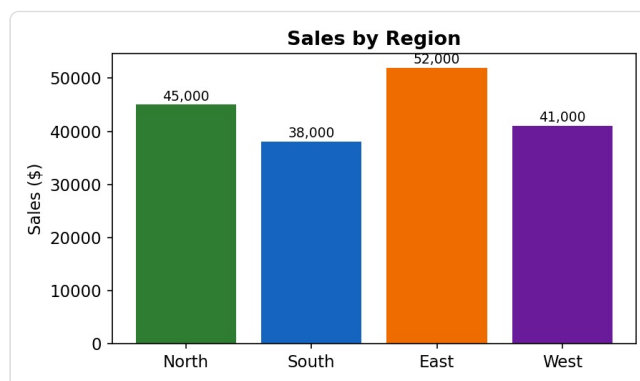
A chart turns rows of numbers into a picture your audience can understand in seconds. Path: select your data → **Insert tab** → **Charts group** → pick a chart type.

Choosing the Right Chart Type

Chart Type	Best for
Column / Bar	Comparing values across categories (e.g. sales by region)
Line	Showing a trend over time (e.g. revenue by month)
Pie	Showing parts of a whole (only when it adds to 100%, ideally ≤ 6 slices)
Clustered Column	Comparing two data series side-by-side (e.g. Q1 vs Q2)

Example 1 — Column Chart: Sales by Region

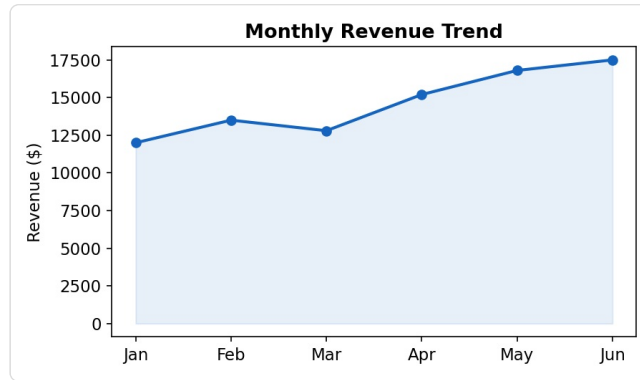
Region	Sales
North	45000
South	38000
East	52000
West	41000



Steps: Select A1:B5 → Insert → Column Chart → 2-D Clustered Column.

Example 2 — Line Chart: Trend Over Time

Month	Revenue
Jan	12000
Feb	13500
Mar	12800
Apr	15200
May	16800
Jun	17500



🔗 **TIP:** Always add a chart **Title** and **axis labels** — Chart Design tab → Add Chart Element. A chart without labels confuses more than it clarifies.

⇒ **PRACTICE:** Enter 4 quarters of revenue for your business idea. Insert a Column chart, then a Line chart from the same data — notice how each tells a slightly different story.

WEEK 3 — Charts & Lookups • DAY 16 OF 30

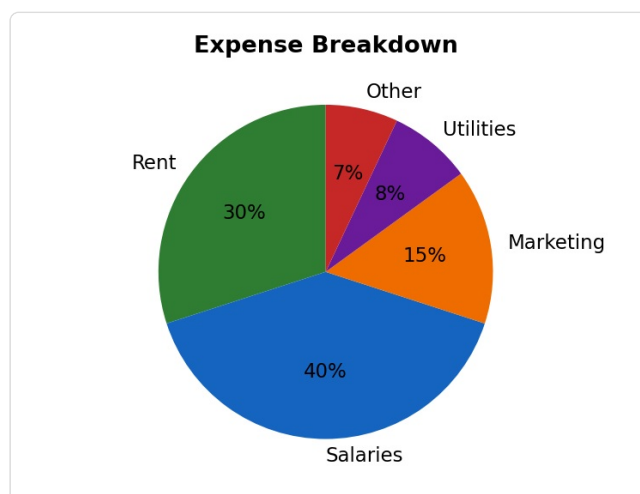
Advanced Charts & Formatting

📌 Teacher's Note

Once students have made one chart, the temptation is to add every visual embellishment Excel offers — 3D effects, extra colors, drop shadows. This is the lesson where I actively teach subtraction rather than addition: a chart usually communicates better with fewer elements, not more. I ask students to remove something from their chart and explain why it still works, which is a harder and more valuable exercise than adding something new.

Example 3 — Pie Chart: Parts of a Whole

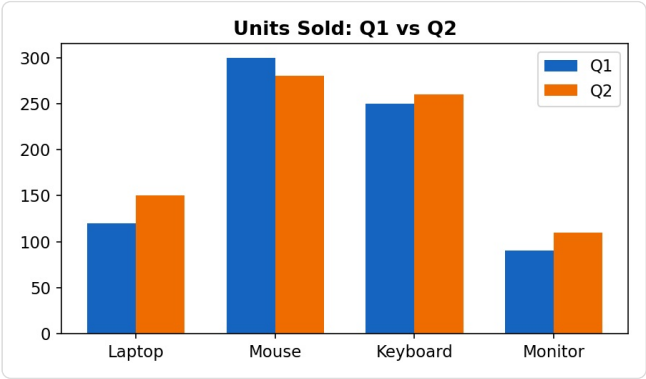
Category	% of Budget
Rent	30
Salaries	40
Marketing	15
Utilities	8
Other	7



⚠ **WATCH OUT:** Never use a pie chart for values that don't add up to a whole, or for more than ~6 categories — it becomes unreadable.

Example 4 — Clustered Column: Comparing Two Series

Product	Q1 Units	Q2 Units
Laptop	120	150
Mouse	300	280
Keyboard	250	260
Monitor	90	110



Formatting a Chart

Task	How
Change chart type	Right-click chart → Change Chart Type
Edit data range	Right-click chart → Select Data
Add data labels	Chart Design → Add Chart Element → Data Labels
Change colors	Chart Design → Change Colors, or click a series and pick Fill Color
Move chart to its own sheet	Right-click chart → Move Chart → New Sheet

Combo Charts

Combine two chart types (e.g. Column for Sales + Line for Target) by selecting your data → Insert → Combo Chart, or Change Chart Type → Combo.

➤ **PRACTICE:** Recreate the expense pie chart with your own 5 budget categories. Then build a clustered column chart comparing planned vs actual spend.

WEEK 3 — Charts & Lookups • DAY 17 OF 30

PivotTables Basics

📌 **Teacher's Note**

PivotTables are, in my experience, the single most intimidating-looking feature in this entire course before students try one, and the single most exciting once they do — because it's the first time a huge, messy dataset visibly resolves into a clean answer in seconds. I spend very little time on the theory and get students dragging fields around almost immediately; PivotTables are one of the rare Excel topics that are genuinely easier to learn by experimenting than by being told how they work.

What Is a PivotTable?

A PivotTable summarizes large datasets instantly — totals, counts, and averages — by dragging fields into a layout, with no formulas required.

Sample Raw Data (Before)

Date	Region	Product	Sales
01-Jun	North	Laptop	45000
02-Jun	South	Mouse	1200
03-Jun	North	Mouse	1500
04-Jun	East	Laptop	52000
05-Jun	South	Laptop	39000
06-Jun	North	Keyboard	2200

Building a PivotTable

1. Click anywhere inside your data
2. **Insert tab → PivotTable → OK** (creates it on a new sheet)
3. In the PivotTable Fields pane, drag fields into the four boxes:

Box	Purpose	Example field
Rows	Categories down the left side	Region
Columns	Categories across the top	Product
Values	The numbers to summarize	Sales (Sum)
Filters	A dropdown to filter the whole table	Date

Result — Total Sales by Region and Product

Region	Laptop	Mouse	Keyboard	Grand Total
North	45000	1500	2200	48700
South	39000	1200		40200
East	52000			52000

Changing the Summary Calculation

By default, PivotTables SUM numbers. Click the field in the Values box → **Value Field Settings** to switch to Count, Average, Max, Min, etc.

 **PRACTICE:** Using 15 rows of sample sales data (Date, Region, Product, Sales), build a PivotTable showing total sales per Region, then switch it to show Average sales per Region.

WEEK 3 — Charts & Lookups • DAY 18 OF 30

PivotTables: Slicers, Grouping & Calculated Fields

 **Teacher's Note**

Slicers and calculated fields turn a static PivotTable into something closer to an interactive tool, and this is usually when non-technical students start describing themselves as "building a dashboard" rather than "making a spreadsheet." That shift in self-perception is worth encouraging — it's often the moment someone decides they're actually good at this.

Slicers — Interactive Filter Buttons

Slicers turn PivotTable filtering into clickable buttons instead of dropdown menus — perfect for dashboards and live classroom demos.

Path: click inside the PivotTable → **PivotTable Analyze → Insert Slicer** → choose a field (e.g. Region).

Grouping Data

Grouping	Use case
Group dates by Month/Quarter/Year	Right-click a date field in the Pivot → Group
Group numbers into ranges	Right-click a number field → Group (e.g. Age: 0-18, 19-35, 36+)

Calculated Fields

Add a custom formula field directly inside the PivotTable.

Path: PivotTable Analyze → Fields, Items & Sets → **Calculated Field**

```
Profit = Sales - Cost // creates a new "Profit" field usable like any other
```

PivotCharts

A PivotChart is a chart linked directly to a PivotTable — when you filter the table (or click a slicer), the chart updates instantly.

Path: PivotTable Analyze → **PivotChart**

Refreshing Data

If your source data changes, the PivotTable does NOT update automatically — right-click it → **Refresh**, or press **Alt+F5**.

⚠ **WATCH OUT:** A common classroom mistake: editing source data and wondering why the PivotTable looks unchanged. Always Refresh after edits.

⇒ **PRACTICE:** On your Day 17 PivotTable, add a Slicer for Product, group any date field by Month, and insert a PivotChart linked to it.

WEEK 3 — Charts & Lookups • DAY 19 OF 30

INDEX & MATCH

📌 Teacher's Note

INDEX and MATCH is consistently the hardest concept in the whole course to teach well, not because the functions are individually complex, but because combining two functions inside one another requires a kind of formula literacy students haven't needed yet. I always teach MATCH completely on its own first, let students see it return a plain position number, and only then wrap it inside INDEX — building the compound formula in two visible stages rather than presenting the finished combination all at once.

Why INDEX & MATCH Beats VLOOKUP

	VLOOKUP	INDEX & MATCH
Direction	Only looks right	Looks left OR right
Speed on huge data	Slower	Faster
If columns are inserted	Can break (col_index_num shifts)	Keeps working

MATCH — Find a Position

```
=MATCH(lookup_value, lookup_array, 0) // 0 = exact match
```

A: Product

Pen

Notebook

Eraser

```
=MATCH("Eraser", A1:A3, 0) // returns 3 (Eraser's position)
```

INDEX — Get a Value by Position

```
=INDEX(range, row_number)
```

```
=INDEX(B1:B3, 3) // returns the 3rd value in column B
```

Combining Them — The Real Power

```
=INDEX(return_range, MATCH(lookup_value, lookup_range, 0))
```

Product	Price
Pen	10
Notebook	60
Eraser	5

```
=INDEX(B2:B4, MATCH("Eraser", A2:A4, 0)) // finds Eraser's row, returns its Price → 5
```

Two-Way Lookup (Row + Column)

	Q1	Q2	Q3
North	45	50	48
South	38	41	39

```
=INDEX(B2:D3, MATCH("South",A2:A3,0), MATCH("Q2",B1:D1,0)) // → 41
```

💡 **TIP:** Think of MATCH as answering "WHERE is it?" and INDEX as answering "WHAT is at that position?" — together they answer any lookup question.

🔗 **PRACTICE:** Rebuild your Day 13 VLOOKUP price lookup using INDEX & MATCH instead. Then try a two-way lookup with a small Region × Quarter sales grid.

WEEK 3 — Charts & Lookups • DAY 20 OF 30

Text Functions: LEFT, RIGHT, MID, TRIM, CONCAT, TEXTJOIN

📌 Teacher's Note

Text functions rarely get the appreciation they deserve because they don't calculate anything — they clean, reshape, and standardize. But in real working life, messy real-world data (inconsistent capitalization, stray spaces, combined fields) is far more common than the tidy example tables we practice on. I tell students plainly: the person who can clean data quickly is often more valuable on a team than the person who can build the fanciest formula.

Common Text Functions

Function	Purpose	Example	Result
LEFT	First N characters	=LEFT("Excel2026",5)	Excel
RIGHT	Last N characters	=RIGHT("Excel2026",4)	2026
MID	Characters from the middle	=MID("Excel2026",2,3)	xce
LEN	Counts characters	=LEN("Excel")	5
TRIM	Removes extra spaces	=TRIM(" Hi there ")	Hi there
UPPER / LOWER	Changes case	=UPPER("excel")	EXCEL
PROPER	Capitalizes each word	=PROPER("john smith")	John Smith

Joining Text: CONCAT and TEXTJOIN

```
=CONCAT(A2," ",B2) // joins First + Space + Last name
```

```
=TEXTJOIN(" ", TRUE, A2:A5) // joins a whole range with " " and skips blanks
```

A: First	B: Last	Full Name
Amit	Sharma	=CONCAT(A2," ",B2) → Amit Sharma

Splitting Text: Text to Columns

Path: **Data tab** → **Text to Columns** — splits one column (e.g. "City, State") into two based on a delimiter like a comma.

Cleaning Messy Data — A Real Example

Raw (Column A)	Cleaned Formula	Result
amit SHARMA	=PROPER(TRIM(A2))	Amit Sharma
excel-2026-course	=SUBSTITUTE(A3,"-"," ")	excel 2026 course

💡 **TIP:** Combine functions! =PROPER(TRIM(A2)) fixes both extra spaces AND inconsistent capitalization in a single formula.

👉 **PRACTICE:** Take a messy list of 5 names with inconsistent spacing/capitalization. Clean them with TRIM + PROPER, then use CONCAT to create email-style usernames like firstname.lastname.

WEEK 3 — Charts & Lookups • DAY 21 OF 30

Week 3 Review + Mini Project

📌 Teacher's Note

The sales dashboard mini-project is deliberately the most open-ended assignment so far — I give requirements, not exact steps. That ambiguity is intentional. Real spreadsheet work rarely comes with a step-by-step recipe, and Week 3's review is where students start practicing the judgment calls (which chart, which layout, what to highlight) that separate someone who knows Excel's features from someone who can actually solve a problem with them.

Week 3 Recap

Day	Skill
15	Column & Line charts
16	Pie & Clustered column charts, formatting
17	PivotTable basics
18	Slicers, grouping, calculated fields
19	INDEX & MATCH
20	Text functions & cleaning data

Mini Project: Sales Dashboard

Using a raw sales dataset (Date, Region, Product, Salesperson, Amount — at least 20 rows):

- ☐ Build a PivotTable: Total Sales by Region and Product
- ☐ Add a Slicer for Salesperson
- ☐ Insert a PivotChart (Clustered Column) linked to the PivotTable
- ☐ Add a Pie chart showing % of total sales by Region
- ☐ In a separate "Lookup" cell, use INDEX & MATCH so typing a Salesperson's name shows their total sales
- ☐ Use TEXTJOIN to build a one-cell summary sentence, e.g. "Top regions: North, East, South"

💡 **TIP:** A one-page dashboard combining a PivotTable, a slicer, and two charts is a strong portfolio piece for students — encourage them to present it out loud.

WEEK 4

Advanced Excel & Automation

Days 22–30: Work with dates, build what-if models, automate with macros, and clean data with Power Query.

Why This Week Matters

The final week is less about new syntax and more about new habits: automating repeated work, planning for errors before they happen, and designing output for someone else to read. These are the skills that separate someone who knows Excel functions from someone colleagues actually want handling the team's spreadsheets. It's also, deliberately, the week with the least hand-holding — real spreadsheet work rarely comes with a tidy set of instructions attached.

This Week's Lessons

DAY 22	Date & Time Functions	DATEDIF, WORKDAY, NETWORKDAYS
DAY 23	Nested Formulas	IFS, AND, OR
DAY 24	What-If Analysis	Goal Seek, Data Tables, Scenarios
DAY 25	Macros & VBA	Recording and running automation
DAY 26	Power Query Basics	Import and clean data
DAY 27	Dashboards	Visualization best practices
DAY 28	Error Handling	IFERROR, ISNA, formula auditing
DAY 29	Shortcuts & Tables	Productivity + teaching tips
DAY 30	Capstone Project	30-day recap and final project

By the End of This Week, You'll Be Able To

- Calculate date differences and business-day deadlines using DATEDIF, WORKDAY, and NETWORKDAYS
- Use Goal Seek and Scenario Manager to test different business assumptions
- Record a simple macro and understand the VBA code it generates
- Import and clean external data using Power Query
- Design a clutter-free dashboard and catch formula errors using IFERROR and formula auditing tools

📌 Teacher's Note

By this point, most of the remaining struggle isn't about understanding any single feature — it's about deciding which tool to reach for first when a real, messy problem shows up with no instructions attached. That's exactly what the capstone project on Day 30 is designed to test, and it's the truest measure of whether the last four weeks actually stuck.

Date & Time Functions

🔑 Teacher's Note

Dates confuse students more than almost any other Excel topic because the underlying mechanism — dates as serial numbers counting from a fixed starting point — is invisible until you go looking for it. Once a student sees that a date is just a number wearing a date-shaped costume, functions like DATEDIF and WORKDAY stop feeling like magic and start feeling like ordinary arithmetic, which is exactly what they are.

| Date & Time Basics

Excel stores dates as serial numbers (Jan 1, 1900 = 1) — that's why you can do math with dates, like subtracting one from another to get a number of days.

| Core Date Functions

Function	Purpose	Example	Result (on 23-Aug-2026)
TODAY()	Current date	=TODAY()	23-Aug-2026
NOW()	Current date + time	=NOW()	23-Aug-2026 14:32
YEAR / MONTH / DAY	Extract parts of a date	=YEAR(A2)	2026
DATEDIF	Difference between 2 dates	=DATEDIF(A2,B2,"d")	Number of days
EDATE	Add/subtract months	=EDATE(A2,3)	Date 3 months later
WORKDAY	Add business days (skips weekends)	=WORKDAY(A2,10)	Date 10 working days later
NETWORKDAYS	Count business days between 2 dates	=NETWORKDAYS(A2,B2)	Working day count

| Worked Example: Employee Tenure

Name	Join Date	Years Worked
Aisha	15-Jun-2019	=DATEDIF(B2,TODAY(),"y") → 7
Ravi	01-Jan-2023	=DATEDIF(B3,TODAY(),"y") → 3

| Worked Example: Project Deadline

```
=WORKDAY(TODAY(), 15) // deadline 15 business days from today, skipping weekends
```

⚠️ **WATCH OUT:** DATEDIF is a hidden/undocumented function (no autocomplete help appears) but it works reliably — the unit codes are "y" (years), "m" (months), "d" (days).

💡 **PRACTICE:** Create a table of 5 employees with Join Dates. Calculate each person's tenure in years using DATEDIF, and use WORKDAY to find a training deadline 20 business days from today.

Nested Formulas: IFS, AND, OR

🔑 Teacher's Note

IFS versus nested IF is a good moment to talk about code readability as its own skill, separate from correctness. Both approaches often produce the identical result, but one is far easier for a colleague — or your future self, six months later — to read and trust. I ask students to write both versions of the same formula and decide which one they'd rather inherit from someone else.

| IFS — Multiple Conditions Without Deep Nesting

IFS checks conditions in order and returns the result for the first TRUE one — cleaner than nested IF for 3+ outcomes.

```
=IFS(B2>=90,"A", B2>=75,"B", B2>=50,"C", TRUE,"F")
```

Compare to the nested IF from Day 9 — same result, easier to read.

Combining AND / OR Inside IFS or IF

```
=IF(AND(B2>=40, C2>=40, D2>=40), "Pass All", "Check Again")

=IF(OR(B2="Late", C2="Late"), "Delay Risk", "On Track")
```

Worked Example: Shipping Cost Rules

Weight (kg)	Cost Formula	Result
2	=IFS(A2<=1,50,A2<=5,100,A2<=10,180,TRUE,250)	100
8	same formula	180
15	same formula	250

Nesting Functions Together

Real spreadsheets often combine several functions in one cell. Example: grade AND a pass/fail flag in one formula:

```
=IF(AVERAGE(B2:D2)>=40, IFS(AVERAGE(B2:D2)>=90,"A",AVERAGE(B2:D2)>=75,"B",TRUE,"C"), "F")
```

 **TIP:** Break complex formulas into helper columns first (e.g. one column for Average, another for Grade) — combine into one cell only once each piece works correctly.

➡ **PRACTICE:** Rebuild your Day 9 grading formula using IFS instead of nested IF. Then add a shipping-cost table with 4 weight tiers using IFS.

WEEK 4 — Advanced • DAY 24 OF 30

What-If Analysis: Goal Seek, Data Tables, Scenarios

 **Teacher's Note**

What-If tools are where Excel starts to feel less like a calculator and more like a decision-support tool, which is a meaningful conceptual shift. Goal Seek in particular tends to delight students because it's the first time they ask Excel a question in reverse — not "what's the answer?" but "what input would I need for a specific answer?" — which mirrors how real planning and budgeting decisions actually get made.

What-If Analysis — 3 Powerful Tools

What-If Analysis (Data tab → What-If Analysis) lets you test different scenarios without rebuilding your spreadsheet.

1. Goal Seek

Goal Seek works backwards: you tell Excel the RESULT you want, and it finds the INPUT needed to get there.

Scenario	Set cell	To value	By changing cell
"What price do I need to charge to hit \$50,000 profit?"	Profit cell	50000	Price cell

Path: Data → What-If Analysis → **Goal Seek**

2. Data Tables

Shows how a result changes across a whole range of input values at once — great for sensitivity analysis (e.g. loan payment at different interest rates).

Interest Rate	Monthly Payment
4%	\$1,432
5%	\$1,528
6%	\$1,627
7%	\$1,728

3. Scenario Manager

Saves several named "what-if" sets of inputs (e.g. "Best Case", "Worst Case", "Most Likely") so you can switch between them instantly.

Scenario	Sales Growth	Cost Increase	Resulting Profit
Best Case	15%	3%	High
Most Likely	8%	5%	Medium
Worst Case	2%	9%	Low

🔗 **TIP:** Goal Seek answers "what input do I need?"; a Data Table answers "what if I try many inputs?"; Scenario Manager answers "what if I compare a few named situations?"

➡ **PRACTICE:** Build a simple loan calculator (Principal, Rate, Years → Monthly Payment using the PMT function). Use Goal Seek to find the interest rate that gives a \$1,500 monthly payment.

WEEK 4 — Advanced • DAY 25 OF 30

Introduction to Macros & VBA

📌 Teacher's Note

Macros are usually the first brush with programming logic many students have ever had, so I treat this lesson gently and with a lot of curiosity rather than rigor. The goal on Day 25 isn't fluency in VBA — it's simply demystifying it, so that seeing a block of code no longer feels intimidating. Watching your own mouse clicks turn into text is a small but genuinely memorable moment for a lot of learners.

What Are Macros?

A macro records a sequence of actions and lets you replay them with one click — perfect for repetitive tasks like formatting a weekly report.

Recording Your First Macro

1. Enable the Developer tab: File → Options → Customize Ribbon → check **Developer**
2. Developer tab → **Record Macro** → give it a name (no spaces) → OK
3. Perform the actions you want recorded (e.g. bold header row, apply currency format)
4. Developer tab → **Stop Recording**
5. Run it again anytime: Developer → Macros → select it → Run

A Peek at VBA (Visual Basic for Applications)

Every recorded macro is written in VBA behind the scenes. View it: Developer → Macros → select → **Edit**.

```
Sub FormatHeader()  
    Range("A1:D1").Font.Bold = True  
    Range("A1:D1").Interior.Color = RGB(46, 125, 50)  
    Range("A1:D1").Font.Color = RGB(255, 255, 255)  
End Sub
```

This simple macro bolds a header row, fills it green, and turns the text white — exactly what recording that sequence of clicks would produce.

Assigning a Macro to a Button

Insert → Shapes (or Form Controls → Button) → draw it → right-click → **Assign Macro** → pick your macro. Now anyone can run it with one click.

⚠ **WATCH OUT:** Macro-enabled workbooks must be saved as .xlsm, not .xlsx, or the macro will be lost.

🔗 **TIP:** For teaching: recording a macro is a gentle, visual introduction to programming logic — students see their clicks turn directly into code.

➡ **PRACTICE:** Record a macro that applies your Day 4 header formatting (bold + fill color) to any range. Save the file as .xlsm and assign the macro to a button.

Power Query Basics: Import & Clean Data

📌 Teacher's Note

Power Query represents a real mindset shift from everything earlier in the course: instead of writing formulas that react to data, you're writing a repeatable *process* that transforms it. I frame it to students as the difference between cleaning a dataset once by hand versus writing down a recipe you can hand to someone else — or rerun yourself next month with a single click.

What Is Power Query?

Power Query is Excel's built-in tool for importing, cleaning, and reshaping data from many sources (CSV, other Excel files, databases, web pages) — without formulas.

Path: **Data tab** → **Get Data** (or Get & Transform Data group)

A Typical Power Query Workflow

1. **Connect** — Get Data → From File / From Web / From Table
2. **Transform** — remove blank rows, rename columns, split columns, change data types, filter rows — all inside the Power Query Editor
3. **Load** — click Close & Load to bring the cleaned result into your worksheet

Common Cleaning Steps

Problem in raw data	Power Query fix
Extra blank rows	Home → Remove Rows → Remove Blank Rows
Inconsistent capitalization	Transform → Format → Capitalize Each Word
Merged/combined column (e.g. "City, State")	Transform → Split Column → By Delimiter
Wrong data type (numbers stored as text)	Click the column type icon → choose correct type
Data from 5 separate files	Get Data → From Folder → Combine

Why It Beats Manual Cleaning

- Every step is recorded — refresh the query and it repeats automatically on new data
- No formulas or helper columns left behind in your sheet
- Handles much larger, messier datasets than manual editing

💡 **TIP:** Think of Power Query as a recipe: once you write the steps (clean this column, split that one), you can re-run the whole recipe on next month's data with one click — Refresh.

🔗 **PRACTICE:** Import a small CSV (or a table you build) into Power Query. Remove any blank rows, split a combined column by a delimiter, and load the cleaned result back into Excel.

Dashboards & Data Visualization Best Practices

📌 Teacher's Note

Dashboard design is the lesson where technical skill and communication skill visibly merge. A technically perfect dashboard that nobody can read in ten seconds has failed at its actual job. I often have students dashboard-review a classmate's work cold, with no explanation allowed, and report back what they understood — it's a humbling and very effective way to learn what "clear" really means in practice.

What Makes a Good Dashboard

- **One page, one purpose** — a viewer should understand the key message in under 10 seconds
- **Most important number first** — top-left is where eyes land first
- **Consistent color meaning** — e.g. green = good, red = needs attention, throughout
- **Minimal clutter** — remove gridlines, unnecessary borders, and 3D chart effects

Dashboard Layout Example

Zone	Content
Top band	Title + key metrics (KPIs) as large numbers
Left/Center	Main chart (trend or comparison)
Right/Bottom	Supporting chart + a filter/slicer

KPI Cards Using Simple Formulas

KPI	Formula
Total Sales	=SUM(Sales)
Avg Order Value	=AVERAGE(Sales)
Orders This Month	=COUNTIFS(Date,">="&DATE(2026,8,1))

Chart Best Practices Recap

Do	Avoid
Use a clear title and axis labels	Unlabeled axes
Sort bar charts by value	Random/alphabetical order when comparing size
Use color to highlight ONE key insight	Rainbow colors with no meaning
Keep pie charts to ≤6 slices	10+ tiny pie slices

Removing Visual Clutter

View tab → uncheck **Gridlines**. Select cells → Home → Borders → No Border. These two small changes instantly make a report look more professional.

🔗 **TIP:** Ask a colleague to look at your dashboard for 5 seconds and tell you the one thing they noticed — if it's not your main point, redesign the layout.

🔗 **PRACTICE:** Take your Day 21 sales dashboard and apply dashboard design principles: remove gridlines, add 2-3 large KPI numbers at the top, and ensure every chart has a title.

WEEK 4 — Advanced • DAY 28 OF 30

Error Handling: IFERROR, ISNA & Formula Auditing

📌 Teacher's Note

Errors are usually treated as failures to avoid, but I teach them as information — every Excel error code is Excel actively telling you something specific about what went wrong, if you know how to read it. Reframing #REF! or #N/A as a diagnostic clue rather than a dead end changes how students troubleshoot for the rest of their working lives, not just in this course.

Common Excel Errors & What They Mean

Error	Meaning	Typical Fix
#DIV/0!	Dividing by zero or an empty cell	Check the denominator; wrap with IFERROR
#N/A	Lookup value not found	Check spelling/spacing; confirm the value exists
#VALUE!	Wrong data type used (e.g. text where a number is expected)	Check for accidental text in a number column
#REF!	Formula refers to a cell that was deleted	Rebuild the reference
#NAME?	Excel doesn't recognize a function name	Check spelling of the function
#####	Column too narrow to display the value	Widen the column (double-click the border)

IFERROR — Catch Errors Gracefully

```
=IFERROR(A2/B2, "Check input") // shows a friendly message instead of #DIV/0!
```

```
=IFERROR(VLOOKUP(A2,Table,3,FALSE), "Not in list")
```

ISNA, ISERROR, ISBLANK — Testing Before Calculating

Function	Tests for
ISNA	Specifically the #N/A error
ISERROR	Any error type
ISBLANK	An empty cell
ISNUMBER / ISTEXT	Whether a cell holds a number or text

Formula Auditing Tools

Path: **Formulas tab** → **Formula Auditing group**

Tool	What it does
Trace Precedents	Draws arrows showing which cells feed INTO this formula
Trace Dependents	Draws arrows showing which formulas USE this cell
Evaluate Formula	Steps through a complex formula piece by piece
Show Formulas (Ctrl+`)	Displays every formula instead of its result, useful for reviewing a whole sheet

➡ **PRACTICE:** Deliberately break a formula (e.g. divide by an empty cell) to see #DIV/0!, then fix it with IFERROR. Use Trace Precedents on one formula in your dashboard to see its data flow.

WEEK 4 — Mastery • DAY 29 OF 30

Keyboard Shortcuts, Excel Tables & Teaching Tips

📌 Teacher's Note

By Day 29, the goal shifts from learning new features to working faster with the ones already learned. Keyboard shortcuts feel like a minor efficiency gain in isolation, but compounded across years of spreadsheet work, they represent a genuinely significant amount of reclaimed time — and, just as importantly, they keep a presenter's hands on the keyboard and their attention on the audience instead of hunting through menus.

Essential Keyboard Shortcuts

Shortcut	Action
Ctrl + C / Ctrl + V	Copy / Paste
Ctrl + Z / Ctrl + Y	Undo / Redo
Ctrl + Arrow key	Jump to the edge of a data region
Ctrl + Shift + L	Toggle filters on/off
Ctrl + 1	Open Format Cells
Ctrl + `	Show/hide all formulas
F2	Edit the active cell
F4	Repeat last action / toggle \$ in a reference while editing
Alt + =	AutoSum
Ctrl + T	Convert range to an official Table
Ctrl + Page Up/Down	Switch between sheets

Excel Tables — An Underused Power Feature

Select your data → **Ctrl+T** to convert it into a formal Table. Benefits:

- Formulas auto-fill down new rows automatically
- Structured references replace cell addresses: `=SUM(Sales[Amount])`

- Built-in filter buttons and banded row styling
- Charts and PivotTables based on a Table auto-expand when you add new rows

Tips for Teaching Excel Effectively

- Use small, realistic datasets (5-15 rows) — big enough to matter, small enough to check by eye
- Have students predict a formula's result BEFORE running it, then verify — this builds real understanding
- Pair "show the formula" (Ctrl+`) with "show the result" so students see both views
- End each session with a 2-minute "what error did you hit, and how did you fix it?" share-out

🔗 **PRACTICE:** Convert one of your practice tables into a Table with Ctrl+T. Add 2 new rows at the bottom and confirm any formulas/formatting extend automatically.

WEEK 4 — Mastery • DAY 30 OF 30

Capstone Project + 30-Day Recap

📌 Teacher's Note

The capstone is intentionally the least-scaffolded assignment in the entire course, because that's the honest test of whether the learning transferred. I remind students — and I'd remind any teacher using this guide — that struggling through the capstone without a clear step-by-step path is not a sign the course failed; it's the actual point. Real spreadsheet problems never arrive with day numbers attached to them.

30-Day Journey Recap

Week	Focus
Week 1	Interface, formulas, references, formatting, basic functions
Week 2	Sorting/filtering, IF logic, COUNTIF/SUMIF, validation, VLOOKUP
Week 3	Charts, PivotTables, INDEX & MATCH, text functions
Week 4	Dates, advanced logic, What-If tools, macros, Power Query, dashboards, error handling

Capstone Project: Complete Business Dashboard

Combine everything from all 30 days into a single polished workbook.

- ☐ **Raw Data sheet:** at least 25 rows (Date, Region, Product, Salesperson, Amount)
- ☐ **Cleaned via Power Query** or TRIM/PROPER formulas
- ☐ **Formulas sheet:** SUM, AVERAGE, COUNTIFS, SUMIFS, IFS grading/status logic
- ☐ **Lookup sheet:** INDEX & MATCH (or VLOOKUP) to search any Salesperson and see their total
- ☐ **Dashboard sheet:** PivotTable + PivotChart, Slicer, at least 2 chart types, KPI numbers at top
- ☐ **Formatting:** conditional formatting on at least one column, gridlines removed, consistent colors
- ☐ **Automation:** one macro (e.g. a "Refresh Dashboard" button)
- ☐ **Error-proofing:** IFERROR wrapped around any lookup formulas

Self-Assessment Rubric (for teaching use)

Criteria	Points
Data is clean and well-organized	20
Formulas are correct and appropriately use \$ references	20
At least 1 PivotTable + 2 charts, clearly labeled	20
Dashboard is visually clear (KPIs, no clutter, consistent colors)	20
Bonus: macro, Power Query, or What-If tool included	20

Where to Go Next

- Explore **Power Pivot** and the **Data Model** for multi-table analysis
- Learn **Power BI** — it shares many concepts with Excel PivotTables and Power Query
- Practice with real public datasets to build a portfolio
- Teach someone else — explaining a concept is the fastest way to master it

 **TIP:** Congratulations on completing the 30-day course! Keep this document as a reference guide — every function and shortcut here can be looked up again whenever you need it.